

Scientific Forgery Image Detection System

CS 3338 Final Project — Group 11

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Jira Project URL:

<https://cs3337-group-11.atlassian.net/jira/software/projects/IFD/boards/167>

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Overview

The Scientific Forgery Image Detection System is a web-based application that detects copy-move forgery in biomedical and scientific images. The system allows users to upload images and receive analysis results identifying possible duplicated regions in scientific figures.

The system comprises:

- A Web Application (Frontend)
- A Backend API Server (Planned)
- A Detection System (Planned)

System Architecture

The application follows this general workflow:

1. Users upload biomedical images through the web interface
2. Frontend sends image to backend API
3. Backend receives and processes the image
4. Detection system analyzes image (future implementation)
5. Backend returns results to frontend
6. User views results in the web interface

Features

Web Application (Frontend)

- Image upload interface
- Image preview
- Submit button for analysis

- Results display section

Backend System (Planned)

- REST API for image upload
- Image handling and validation
- JSON responses
- Communication with detection module

Detection System (Planned)

- Copy-move forgery detection
- Identification of duplicated regions
- Image region comparison
- Pixel-level analysis

Technologies Used

Category	Technologies
Frontend	HTML
Backend	Python, FastAPI, Uvicorn
Planned Libraries	OpenCV, NumPy, Pillow, PyTorch